## Solutions

**Question 3.1**: Use the chain rule to differentiate the following functions:

(a) Split the function and differentiate:

$$\begin{aligned} u(v) = \exp(v), \quad &\Longrightarrow \quad \frac{du}{dv} = \exp(v), \\ v(x) = x^2 + 5x + 2 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x + 5 \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= (2x+5)\exp(x^2+5x+2) \end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(v) = \cos(v), \quad &\Longrightarrow \quad \frac{du}{dv} = -\sin(v), \\ v(x) = 3x + 5 \quad &\Longrightarrow \quad \frac{dv}{dx} = 3 \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= -3\sin(3x+5) \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(v) = v^2, \quad &\Longrightarrow \quad \frac{du}{dv} = 2v, \\ v(x) = \sin(x) \quad &\Longrightarrow \quad \frac{dv}{dx} = \cos(x) \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= 2\sin(x)\cos(x) \end{aligned}$$

**Question 3.2**: Use the product rule to differentiate the following functions:

(a) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \exp(x) \quad &\Longrightarrow \quad \frac{dv}{dx} = \exp(x) \end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \exp(x) + x\exp(x) \\ &= (1+x)\exp(x)\end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(x) = x^2, \quad &\Longrightarrow \quad \frac{du}{dx} = 2x, \\ v(x) = \exp(4x) \quad &\Longrightarrow \quad \frac{dv}{dx} = 4\exp(4x)\end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= 4x^2\exp(4x) + 2x\exp(4x) \\ &= (2x+4x^2)\exp(4x)\end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(x) = x^2 + 9, \quad &\Longrightarrow \quad \frac{du}{dx} = 2x, \\ v(x) = \exp(-x) \quad &\Longrightarrow \quad \frac{dv}{dx} = -\exp(-x)\end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= 2x\exp(-x) - (x^2+9)\exp(-x) \\ &= -(x^2 - 2x + 9)\exp(-x)\end{aligned}$$

**Question 3.3**: Use the quotient rule to differentiate the following functions:

(a) Split the function and differentiate:

$$\begin{aligned} u(x) = \exp(-x), \quad &\Longrightarrow \quad \frac{du}{dx} = -\exp(-x), \\ v(x) = x \quad &\Longrightarrow \quad \frac{dv}{dx} = 1\end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{-x\exp(-x) - \exp(-x)}{x^2} \\ &= -\frac{(1+x)\exp(-x)}{x^2}\end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = x^2 + 25 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{x^2 + 25 - 2x^2}{(x^2+25)^2} \\ &= \frac{25 - x^2}{(x^2+25)^2} \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(x) = x^2, \quad &\Longrightarrow \quad \frac{du}{dx} = 2x, \\ v(x) = x^3 + 25x \quad &\Longrightarrow \quad \frac{dv}{dx} = 3x^2 + 25 \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{2x(x^3+25x) - x^2(3x^2+25)}{(x^3+25x)^2} \\ &= \frac{2x^4 + 50x^2 - 3x^4 - 25x^2}{(x^3+25x)^2} \\ &= \frac{25x^2 - x^4}{(x^3+25x)^2} \end{aligned}$$

**Question 3.4**: Use the chain rule for (a) and the result of (a) in (b)-(e) find $\frac{dy}{dx}$ of the following equations (Hint: you may need to use a mixture of chain, product and quotient rules):

(a) Split the function and differentiate:

$$\begin{aligned} u(v) = \sqrt{v}, \quad &\Longrightarrow \quad \frac{du}{dv} = \frac{1}{2\sqrt{v}}, \\ v(x) = x^2 + 25 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= \frac{2x}{2\sqrt{x^2+25}} \\ &= \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(v) = \sin(v), \quad &\Longrightarrow \quad \frac{du}{dx} = \cos(v), \\ v(x) = \sqrt{x^2+25} \quad &\Longrightarrow \quad \frac{dv}{dx} = \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= \frac{x}{\sqrt{x^2+25}} \cos(\sqrt{x^2+25}) \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \sqrt{x^2+25} \quad &\Longrightarrow \quad \frac{dv}{dx} = \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

Apply the product rule:

$$\begin{aligned} \frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \sqrt{x^2+25} + \frac{x^2}{\sqrt{x^2+25}} \end{aligned}$$

(d) Split the function and differentiate:

$$\begin{aligned} u(x) = x, \quad &\Longrightarrow \quad \frac{du}{dx} = 1, \\ v(x) = \sqrt{x^2+25} \quad &\Longrightarrow \quad \frac{dv}{dx} = \frac{x}{\sqrt{x^2+25}} \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} = \frac{\sqrt{x^2+25} - \frac{x^2}{\sqrt{x^2+25}}}{x^2+25} \\ &= \frac{\sqrt{x^2+25}}{x^2+25} - \frac{\frac{x^2}{\sqrt{x^2+25}}}{x^2+25} = \frac{\sqrt{x^2+25}}{x^2+25} - \frac{x^2}{\sqrt{x^2+25}} \times \frac{1}{x^2+25} \\ &= \frac{1}{\sqrt{x^2+25}} - \frac{x^2}{(x^2+25)^{\frac{3}{2}}} \end{aligned}$$

**Question 3.5**:

(a) Split the function and differentiate:

$$\begin{aligned} u(t) = 4 - t^2, \quad &\Longrightarrow \quad \frac{du}{dt} = -2t, \\ v(x) = 1 + 5t^2 \quad &\Longrightarrow \quad \frac{dv}{dt} = 10t \end{aligned}$$

Apply the product rule:

$$\begin{aligned}\frac{dy}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= 10t(4-t^2) - 2t(1+5t^2) \\ &= 2t(19-10t^2)\end{aligned}$$

Next, we need to know where the function is not changing and so we set the derivative equal to zero and solve. From the factored form of the derivative we can see that the derivative will be zero at:

$$t = 0\ , t = \pm\sqrt{\frac{19}{10}} = \pm 1.3784$$

We need to know is where the derivative is positive (and hence the function is increasing) or negative (and hence the function is decreasing). Because the derivative is continuous we know that the only place it can change sign is where the derivative is zero. We have been given 3 point where the derivative is zero. Lets use these points to define some regions to evaluate the sign of the derivative. This is best illustrated using Figure 3.1. From this we get the following increasing/decreasing information.

$$\begin{aligned}\textit{Increasing}: &\quad -\infty < t < -\sqrt{\frac{19}{10}}, \quad 0 < t < \sqrt{\frac{19}{10}} \\ \textit{Decreasing}: &\quad -\sqrt{\frac{19}{10}} < t < 0, \quad \sqrt{\frac{19}{10}} < t < \infty\end{aligned}$$

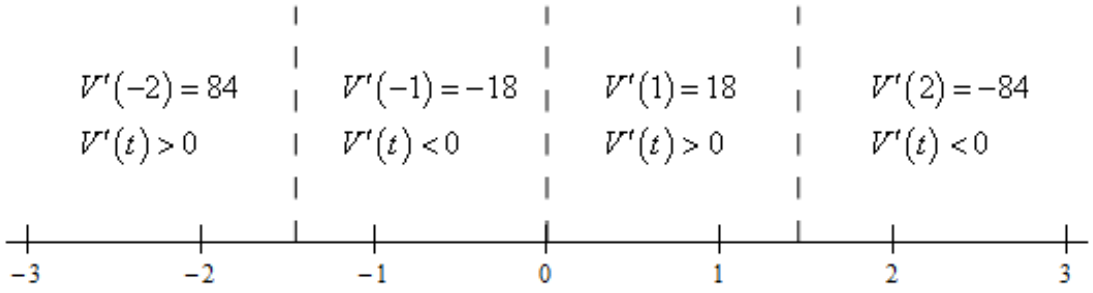



**Figure 3.1:** Here we identify where along the $x$ axis the function derivative has a value of 0 [dashed lines] and then interrogate the sign of the derivatives either side. From this information we can identify regions where the function is increasing or decreasing.

(b) Split the function and differentiate:

$$\begin{aligned}u(x) = x - x^2, \quad &\Longrightarrow \quad \frac{du}{dx} = 1 - 2x, \\ v(x) = 1 + 8x^2 \quad &\Longrightarrow \quad \frac{dv}{dx} = 16x\end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{v \times \frac{du}{dx} - u \times \frac{dv}{dx}}{v^2} \\ &= \frac{(1-2x)(1+8x^2) - 16x(x-x^2)}{(1+8x^2)^2} \\ &= \frac{1-2x-8x^2}{(1+8x^2)^2}\end{aligned}$$

Next, we need to know where the function is not changing and so we set the derivative equal to zero and solve. The denominator will never be zero for any real number and so the derivative will only be zero where the numerator is zero. Thus:

$$1-2x-8x^2 = 0$$

When we solve this we arrive at the following roots:

$$x = -\frac{1}{2}, x = \frac{1}{4}$$

We need to know is where the derivative is positive (and hence the function is increasing) or negative (and hence the function is decreasing). Because the derivative is continuous we know that the only place it can change sign is where the derivative is zero. We have been given 3 point where the derivative is zero. Lets use these points to define some regions to evaluate the sign of the derivative. This is best illustrated using the Figure 3.2.From this we get the following increasing/decreasing information.

$$\begin{aligned}\textit{Increasing}: &\quad -\frac{1}{2} < x < \frac{1}{4} \\ \textit{Decreasing}: &\quad -\infty < x < -\frac{1}{2}, \quad \frac{1}{4} < x < \infty\end{aligned}$$

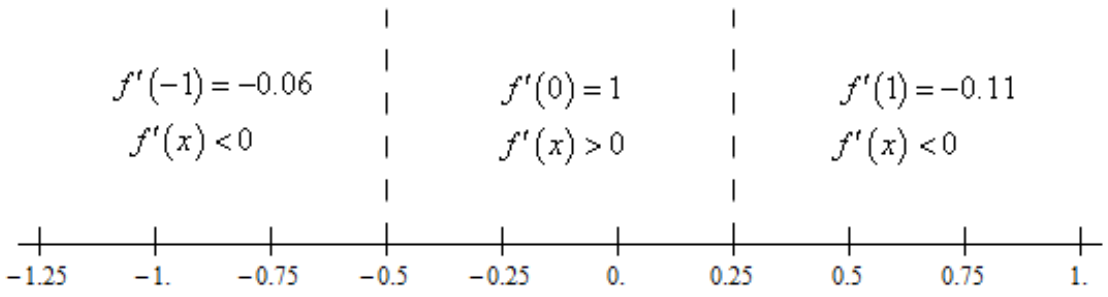



**Figure 3.2:** Here we identify where along the $x$ axis the function derivative has a value of 0 [dashed lines] and then interrogate the sign of the derivatives either side. From this information we can identify regions where the function is increasing or decreasing.

**Question 3.6**:

(a) In this Question 3.we need to apply the chain rule twice. To make sure we don't make any mistakes, lets do it part by part. First we differentiate $y = \sin(x^2)$. Split the function and differentiate:

$$\begin{aligned} u(v) = \sin(v), \quad &\Longrightarrow \quad \frac{du}{dv} = \cos(v), \\ v(x) = x^2 \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dy}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= 2x\cos(x^2) \end{aligned}$$

Now lets apply the chain rule to $Z = (\sin(x^2))^2$. Split the function and differentiate:

$$\begin{aligned} u(v) = v^2, \quad &\Longrightarrow \quad \frac{du}{dv} = 2v, \\ v(x) = \sin(x^2) \quad &\Longrightarrow \quad \frac{dv}{dx} = 2x\cos(x^2) \end{aligned}$$

Apply the chain rule:

$$\begin{aligned} \frac{dZ}{dx} &= \frac{du}{dv} \times \frac{dv}{dx} \\ &= 2sin(x^2) \times 2x\cos(x^2) \\ &= 4xsin(x^2)\cos(x^2) \end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(x) = 1 + \sqrt{x^3} = 1 + x^{\frac{3}{2}}, \quad &\Longrightarrow \quad \frac{du}{dx} = \frac{3}{2}x^{\frac{1}{2}}, \\ v(x) = x^{-3} - 2\sqrt[3]{x} = x^{-3} - 2x^{\frac{1}{3}} \quad &\Longrightarrow \quad \frac{dv}{dx} = -3x^{-4} - \frac{2}{3}x^{-\frac{2}{3}} \end{aligned}$$

Apply the product rule:

$$\begin{aligned} \frac{dS}{dx} &= u \times \frac{dv}{dx} + v \times \frac{du}{dx} \\ &= \left(\frac{3}{2}x^{\frac{1}{2}}\right)\left(x^{-3} - 2x^{\frac{1}{3}}\right) + \left(1 + x^{\frac{3}{2}}\right)\left(-3x^{-4} - \frac{2}{3}x^{-\frac{2}{3}}\right) \\ &= -3x^{-4} - \frac{3}{2}x^{-\frac{5}{2}} - \frac{2}{3}x^{-\frac{2}{3}} - \frac{11}{3}x^{\frac{5}{6}} \end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned} u(w) = 3w + w^4, \quad &\Longrightarrow \quad \frac{du}{dw} = 3 + 4w^3, \\ v(w) = 2w^2 + 1 \quad &\Longrightarrow \quad \frac{dv}{dw} = 4w \end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}\frac{dy}{dw} &= \frac{v \times \frac{du}{dw} - u \times \frac{dv}{dw}}{(2w^2+1)^2} \\ &= \frac{(2w^2+1)(3+4w^3) - 4w(3w+w^4)}{(2w^2+1)^2} \\ &= \frac{4w^5+4w^3-6w^2+3}{(2w^2+1)^2}\end{aligned}$$

## Real-world problems

**Question 3.7**: Imagine we are studying the volume of water in an ecosystem reservoir over time (in hours).

(a) Split the function and differentiate:

$$\begin{aligned} u(v) = V_0 \exp(v), \quad &\Longrightarrow \quad \frac{du}{dv} = V_0 \exp(v), \\ v(t) = -\frac{t^2}{600} \quad &\Longrightarrow \quad \frac{dv}{dt} = -\frac{t}{300}\end{aligned}$$

Apply the chain rule:

$$\begin{aligned}\frac{dV}{dt} &= \frac{du}{dv} \times \frac{dv}{dt} \\ &= -\frac{V_0 \times t}{300} \exp\left(-\frac{t^2}{600}\right)\end{aligned}$$

The drainage rate at $t = 10$ is

$$\begin{aligned}\frac{dV}{dt}(10) &= -\frac{10 \times 10}{300} \exp\left(-\frac{10^2}{600}\right) \\ &= -\frac{1}{3} \exp\left(-\frac{1}{6}\right) \\ &= -0.1829\end{aligned}$$

(b) Split the function and differentiate:

$$\begin{aligned} u(t) = \frac{V_0(t+10)}{10}, \quad &\Longrightarrow \quad \frac{du}{dt} = \frac{V_0}{10}, \\ v(t) = \exp\left(-\frac{t}{5}\right) \quad &\Longrightarrow \quad \frac{dv}{dt} = -\frac{1}{5} \exp\left(-\frac{t}{5}\right)\end{aligned}$$

Apply the product rule:

$$\begin{aligned}
\frac{dV}{dt} &= u \times \frac{dv}{dt} + v \times \frac{du}{dt} \\
&= \frac{V_0(t+10)}{10} \times -\frac{1}{5}\exp\left(-\frac{t}{5}\right) + \exp\left(-\frac{t}{5}\right) \times \frac{V_0}{10} \\
&= -\frac{V_0(t+10)}{50}\exp\left(-\frac{t}{5}\right) + \frac{V_0}{10}\exp\left(-\frac{t}{5}\right) \\
&= V_0 \exp\left(-\frac{t}{5}\right)\left(-\frac{t+10}{50} + \frac{1}{10}\right) \\
&= V_0 \exp\left(-\frac{t}{5}\right)\left(\frac{5-t-10}{50}\right) \\
&= -\frac{V_0(t+5)}{50}\exp\left(-\frac{t}{5}\right)
\end{aligned}$$

The drainage rate at $t = 10$ is

$$\begin{aligned}
\frac{dV}{dt}(10) &= -\frac{10 \times (10+5)}{50}\exp\left(-\frac{10}{5}\right) \\
&= -3\exp(-2) \\
&= -0.406
\end{aligned}$$

(c) Split the function and differentiate:

$$\begin{aligned}
u(t) = t + 10, \quad &\Longrightarrow \quad \frac{du}{dt} = 1, \\
v(t) = t^2 + 10 \quad &\Longrightarrow \quad \frac{dv}{dt} = 2t
\end{aligned}$$

Apply the quotient rule:

$$\begin{aligned}
\frac{dV}{dt} &= V_0 \frac{v \times \frac{du}{dt} - u \times \frac{dv}{dt}}{(2w^2+1)^2} \\
&= V_0 \frac{t^2 + 10 - 2t \times (t+10)}{(t^2+10)^2} \\
&= V_0 \frac{-t^2 - 20t + 10}{(t^2+10)^2}
\end{aligned}$$

The drainage rate at $t = 10$ is

$$\begin{aligned}
\frac{dV}{dt}(10) &= 10 \times \frac{-10^2 - 20 \times 10 + 10}{(10^2+10)^2} \\
&= 10 \times \frac{-100 - 200 + 10}{110^2} \\
&= -0.2397
\end{aligned}$$